Statistics Before Programs: Block ℓ_p-Norms for Cyclic Join Bounds

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***Abstract*—Pessimistic cardinality estimation is usually advanced by strengthening the *program* that combines degree statistics, from fractional edge covers to the entropic linear program over the Shannon cone. Replacing a max-degree product with a fractional cover LP is worth $654\times$ on cycle-8, but on single-relation statistics it is close to the last step: on a fixed hyperpath our cover LP is already optimal over every valid weight vector, yet sits 1.25–$1.43\times$ (synthetic) and a median 2.5–$4.3\times$ (real) above an entropic optimum that LpBound's flow LP reaches in polynomial time. The remaining headroom lies in *which statistics are materialized*—and richer statistics can reopen the program.**

We separate the two axes with LpH-CEG, which combines multi-way block extensions with ℓ_p-norm degree sequences. Block statistics—degree sequences of two-relation joins conditioned on their interfaces—are worth a median $56\times$ on short cycles of a pre-registered Hetionet suite (3K-row samples), 80–$2{,}500\times$ on longer ones ($|V|$=6–10) and 4–$43\times$ (17–$177\times$ with the flow LP) on a registered IMDb suite from JOB's snapshot, below the entropic LpBound on all 349 cycle queries over real data, with no violation among the 236 that have an exact count. The entropic program still wins on those same statistics, by 1.5–$3.8\times$ (synthetic) and a median 3.2–$5.1\times$ (real): we change what it is given, not the program. That strongest cell need not cost $2^{|V|}$: extending Im et al.'s flow LP to ℓ_p block constraints reaches the polymatroid bound over block statistics in polynomial time on $12/12$ synthetic instances, every synthetic even cycle up to k=10 and $189/240$ real-data queries. Under that program block statistics add at least a median $63\times$ on Hetionet beyond pair-join sizes (a proxy for CorrBound's two-relation statistic); and for a block of two binary relations all but one of them need no join—the ℓ_2 norm of its common-neighbour counts, which restores the full bound on $85/120$ Hetionet and $116/120$ IMDb queries. We prove an ℓ_p Composition Inequality for validity—the redundant coverage behind every cyclic gain acts as a fractional cover over the polymatroid cone—and report the operational limits plainly.

***Index Terms*—cardinality estimation, cyclic queries, ℓ_p-norms, fractional cover, degree sequences, pessimistic estimation**

I. Introduction

Cardinality estimation—predicting the output size of a join query before execution—is the most critical input for query optimization. Errors compound multiplicatively across join operators, and cyclic queries are the hardest case: the constraints imposed by cycle-closing edges break independence assumptions that underpin most estimators.

The information-theoretic approach to cardinality bounds has seen rapid progress. The AGM bound [1] uses fractional edge covers; PANDA [2] and the degree-based MOLP bound [3] add max-degree (ℓ_∞) constraints; and most recently, **LpBound** [4] generalizes to all ℓ_p-norms of degree sequences, achieving orders-of-magnitude improvements on skewed data via an LP over entropy inequalities. Independently, multi-way *block extensions* have been proposed to capture joint constraints across multiple relations [5], [6].

However, a key gap remains: *no existing method exploits the structural redundancy that cyclic queries create in the fractional cover constraint.* LpBound operates on single-relation extensions only and does not characterize which query structures benefit from LP optimization. Block extension methods use only max-degree statistics.

a) Our contribution.: LpH-CEG bridges ℓ_p-norm LP optimization with multi-way block decompositions through a *fractional block cover LP*: on cyclic queries an attribute is covered by several steps, and the slack admits fractional weights $w_j < 1$ that, with intermediate norms (ℓ_2), tighten the max-degree bound (w=1, p=∞) dramatically. LpBound's entropic LP over the Shannon cone is related precisely in Section IV-A. Figure 1 places the bs=1 configuration in the design space: near the entropic optimum and decomposable into per-step residuals, but—since LpBound's flow LP reaches that optimum in polynomial time—not cheaper in any sense that matters. The thesis is at the other end: at $bs\geq 2$ the statistics cost more to collect, once per database, and the bound is far tighter (Section VI-C).

Our contributions are:

1) **On single-relation statistics the inference axis is closed (Section IV-A).** On a fixed hyperpath the cover LP is optimal over *every* valid weight vector: a cutting-plane search adding Shannon cuts to a relaxation converges back to exactly the cover bound. The gap to the entropic optimum comes from statistics the hyperpath leaves unused, not from the weighting—and LpBound's flow LP, which uses them, reaches that optimum in polynomial time, so at bs=1 the cover LP is dominated as a bound.
2) **The statistics axis is open (Section VI-C).** Block statistics $\mu_p(F \mid I)$, $|F| \geq 2$, are worth a median $56\times$ on the 109 short cycles of a pre-registered Hetionet suite (rows capped at 3K) and 80–$2{,}500\times$ on $120/120$ longer Hetionet cycles ($|V|$=6–10), against a median 2.5–$3.6\times$ of inference headroom. On 120 IMDb cycles from JOB's snapshot the margin is smaller: 4–$43\times$ with the cover LP (17–$177\times$ with the flow LP) against 3.8–$4.3\times$, level at $|V|$=6. On homogeneous synthetic cycles the axes are comparable.

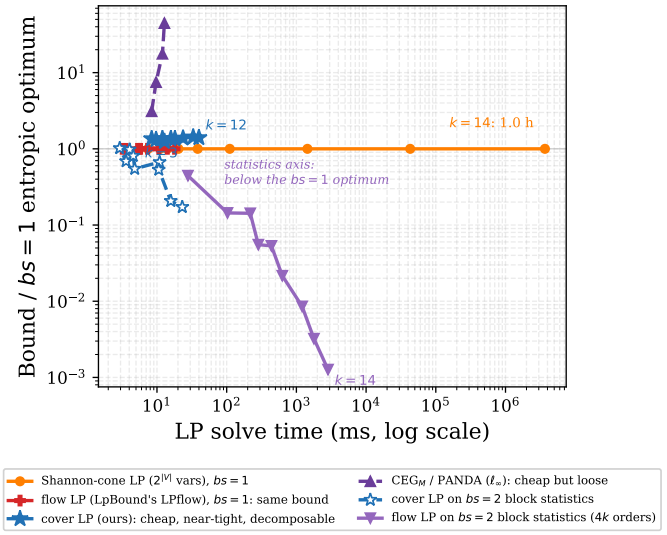


Fig. 1. Both axes on one *measured* cost–tightness plot (each point a synthetic cycle-k query; y is relative to the entropic optimum on single-relation statistics). Above y=1, the inference axis at bs=1: CEG$_M$/PANDA is cheap but up to 45× loose, the cover LP within 1.3× (7–46 ms end to end), and the optimum itself 43 s at k=12 as a $2^{|V|}$ program but a 3–18 ms solve ($k\leq$14) as LpBound's flow LP—that axis is closed. Below y=1, the statistics axis: the same programs on bs=2 block statistics, the cover LP level with the optimum at $k\leq$4 and up to 6× below it, the flow LP (Section VI-E) 2–800× below. Data: `scale_entropic.csv`, `flow_scale.csv`, `baseline_comparison.csv`.

3) **Richer statistics reopen the program (Sections VI-C, VI-E).** On the same block statistics the polymatroid optimum beats the cover LP by 1.5–3.8× (12/12 synthetic) and a median 3.2–5.1× (Hetionet), against 1.15–1.5× on single-relation statistics; on IMDb it stays level at 200 films (3.3–4.2×) and grows to 6.8× at 1,000. Extending Im et al.'s flow LP to ℓ_p block constraints reaches that optimum in polynomial time on 12/12 synthetic instances, every synthetic even cycle to k=10 and 189/240 real-data queries (0.03–4.4 s at k=3–16).
4) **Which block statistic matters (Section VI-C).** Under the same program (the order-relaxed flow LP, so a lower bound on the gain), block statistics carry a further median 63× on Hetionet beyond pair-join sizes, a proxy for the two-relation statistic CorrBound adds (itself worth 31× there; on IMDb, where half the pairs are self-joins, sizes add almost nothing). For a block of two binary relations all but one of them come from single-relation passes; the exception, the ℓ_2 norm of the common-neighbour counts, restores the full bound on 85/120 Hetionet and 116/120 IMDb queries.
5) **Validity, cost and limits (Sections IV, VI, V).** An ℓ_p Composition Inequality makes the cover bound valid for every hyperpath, block size and p; the proof turns redundant coverage into a fractional cover, and 207K databases plus an LP certificate over 6,197 hyperpath structures independently agree. Block statistics cost a median 45× more to collect, once per database; cataloged on Hetionet, a bs=2 estimate costs 10 ms. Guiding join order, the bound trims the catastrophic tail on synthetic cliques (P95 slowdown 1.2× vs. 5.8×) rather than the average, and on IMDb block statistics pick the exact-count order on 24/40 queries, yet those left-deep plans still lose to DuckDB's own; and a per-step residual ordering heuristic does not generalize (median −1.3% over 66 queries).

II. Preliminaries

A. Query Hypergraphs

A conjunctive query $\mathcal{Q}$ over relations $R_1, \ldots, R_m$ with attributes $V = \{a_1, \ldots, a_n\}$ is represented as a hypergraph $H = (V, \mathcal{E})$ where each hyperedge $e_i \in \mathcal{E}$ corresponds to the attribute set of relation R_i. The full natural join is $\mathcal{Q} = \bowtie_{i=1}^{m} R_i$.

B. CEG and MOLP

Cardinality Estimation Graphs [6], [7] model both optimistic and pessimistic estimators as path-finding in a directed graph. The pessimistic CEG$_M$ uses max-degree edge weights: for relation R with interface I, $\mu(R \mid I) = \max_{a_I} |\text{Ext}_R(a_I)|$. The MOLP bound [3] equals the minimum-weight path in CEG$_M$.

C. Block Extensions and Hyperpaths

A *block* $F \subseteq \mathcal{E}$ is a connected subset of relations. The *block extension factor* [5] is $\mu(F \mid I) = \max_{a_I} |\text{Ext}_F(a_I)|$, where $I = \text{attr}(S) \cap \text{attr}(F)$ is the interface with the current state. A *hyperpath* $P = (S_0{=}\emptyset, F_1, S_1, \ldots, F_m, S_m{=}\mathcal{E})$ decomposes the query into sequential block extensions. The product $\prod_{j=1}^{m} \mu(F_j \mid I_j)$ is a valid cardinality upper bound (Composition Lemma [5]). Our framework generalizes this product by replacing ℓ_∞ norms with LP-optimized ℓ_p-norms.

D. LpBound: ℓ_p-Norm Degree Sequences

LpBound [4] uses ℓ_p-norms of degree sequences as tighter statistics. For relation R with degree sequence $\mathbf{d} = (d_1, \ldots, d_k)$ where $d_i = |\text{Ext}_R(a_i)|$, the ℓ_p-norm is $\|\mathbf{d}\|_p = (\sum_i d_i^p)^{1/p}$. The core inequality is: for any $p \geq 1$,

$$\tfrac{1}{p}\, h(U) + h(V \mid U) \leq \log \|\mathbf{d}\|_p$$

where h denotes entropy. Optimizing over all valid inequalities via an LP yields bounds orders of magnitude tighter than max-degree alone.

III. LpH-CEG Framework

A. ℓ_p-Norm Block Extension Factor

Definition 1 (ℓ_p Block Extension Factor)**.** *For a block F with interface I, let $\mathbf{d} = (d_1, \ldots, d_k)$ be the degree sequence where $d_i = |Ext_F(a_i)|$ for each distinct interface value $a_i \in \pi_I(\mathcal{Q}_F)$. The ℓ_p block extension factor is:*

$$\mu_p(F \mid I) = \|\mathbf{d}\|_p = \Big(\sum_{i=1}^{k} d_i^p\Big)^{1/p}$$

Special cases: $\mu_\infty(F \mid I) = \max_i d_i = \mu(F \mid I)$ (max-degree); $\mu_1(F \mid I) = \sum_i d_i = |\mathcal{Q}_F|$ (cardinality).

B. Fractional Block Cover LP

Given a hyperpath P with m steps, we define a variable $w_{j,p} \geq 0$ for each step j and each p-value. The LP minimizes the log-space bound:

$$\min \sum_{j=1}^{m} \sum_{p \in \mathcal{P}} w_{j,p} \cdot \log_2 \mu_p(F_j \mid I_j) \tag{1}$$

subject to the fractional block cover constraints: for each attribute $a \in V$,

$$\sum_{p \in \mathcal{P}} \left(\sum_{j:\, a \in \text{attr}(F_j) \setminus I_j} w_{j,p} \; + \; \sum_{j:\, a \in I_j} \frac{w_{j,p}}{p} \right) \; \geq \; 1 \tag{2}$$

The LP has $O(m \cdot |\mathcal{P}|)$ variables and $|V|$ constraints, solvable in one to two milliseconds for typical queries.

C. Why Cyclic Queries Benefit

Proposition 2 (Acyclic rigidity)**.** *For an acyclic hyperpath with binary extensions ($|F_j| = 1$), every attribute is new at exactly one step. No attribute is then covered twice as an* extension*, so no step's weight can be reduced by another step's extension term, and on star queries the LP returns $w_j = 1$ throughout, recovering the max-degree bound. It does not follow that every acyclic weight is ≥ 1: a later step may cover an attribute through its* interface *at $p = 1$, and on a 4-path the LP returns $(1, 0, 1)$. See Section VI-I for what this does and does not buy.*

Proposition 3 (Cyclic slack)**.** *In a cycle of length k with binary extensions, the cycle-closing attribute appears in ≥ 2 steps (as extension in one, interface in another). The cover constraint allows distributing weight across these steps, enabling $w_j < 1$ and intermediate ℓ_p-norms.*

D. Algorithm

Algorithm 1 summarizes the LpH-CEG estimation pipeline.

The LP in line 10 has $O(m \cdot |\mathcal{P}|)$ variables ($m \leq n$ steps, $|\mathcal{P}| = 3$ in practice) and n constraints. We solve it via HiGHS [8] in < 1 ms for all queries in our experiments. Line 13 ensures LpH-CEG never produces a looser bound than the standard max-degree bound.

Algorithm 1 LpH-CEG Estimation

Require: Query hypergraph H, relation data D, block size limit b, p-values $\mathcal{P} = \{p_1, \ldots, p_r\}$
Ensure: Upper bound $\hat{C}$ on $|\mathcal{Q}(D)|$
1: Enumerate blocks $\mathcal{B} \leftarrow \{F \subseteq \mathcal{E} : |F| \leq b,\ F \text{ connected}\}$
2: **for** each block $F \in \mathcal{B}$ and interface I **do**
3: Compute join $Q_F \leftarrow \bowtie_{R \in F} R$
4: Collect degree sequence $\mathbf{d}_{F,I}$
5: Compute $\mu_p(F \mid I)$ for each $p \in \mathcal{P}$
6: **end for**
7: Build block-extension DAG with ℓ_∞ weights
8: Find K shortest hyperpaths $P_1, \ldots, P_K$ $\{K{=}1$: default$\}$
9: **for** each hyperpath $P_i = (F_1, \ldots, F_m)$ **do**
10: Solve LP (1)–(2) for weights $(w^*_{j,p})$
11: $\hat{C}_i \leftarrow \prod_{j,p} \mu_p(F_j \mid I_j)^{\,w^*_{j,p}}$
12: **end for**
13: **return** $\min(\hat{C}_1, \ldots, \hat{C}_K,$ max-deg bound$)$

IV. Theoretical Results

Proofs not given in full here are in the extended version shipped with the artifact.

Theorem 4 (ℓ_p Composition Inequality)**.** *Let $P = (S_0{=}\emptyset, \ldots, S_m{=}\mathcal{E})$ be a hyperpath, and let $(w_{j,p})$ satisfy the cover constraint* (2)*. Then:*

$$|\mathcal{Q}| \leq \prod_{j=1}^{m} \prod_{p \in \mathcal{P}} \mu_p(F_j \mid I_j)^{\,w_{j,p}}$$

Proof. Let h be the entropy of the uniform distribution on $\mathcal{Q}(D)$, so $h(V) = \log_2 |\mathcal{Q}|$. Write $A_j = \text{attr}(F_j)$, $V_j = \text{attr}(S_j)$ and $E_j = A_j \setminus I_j$: the E_j partition V and $I_j \subseteq V_{j-1}$. Since $\pi_{A_j} \mathcal{Q} \subseteq \bowtie_{R \in F_j} R$, the ℓ_p inequality of [9] gives $\frac{1}{p} h(I_j) + h(E_j \mid I_j) \leq \log_2 \mu_p(F_j \mid I_j)$. Weight by $w_{j,p}$ and sum; with $w_j = \sum_p w_{j,p}$ and $s_j = \sum_p w_{j,p}/p$ it remains to show $\sum_j [s_j h(I_j) + w_j h(E_j \mid I_j)] \geq h(V)$, where (2) reads $w_{j(a)} + \sum_{j' : a \in I_{j'}} s_{j'} \geq 1$ for the step $j(a)$ that binds a. Let $g_j = h(E_j \mid V_{j-1})$ and $\delta_j = \max(0, 1 - w_j)$. (i) By the chain rule and submodularity, $h(I_{j'}) \geq \sum_j h(I_{j'} \cap E_j \mid V_{j-1})$. (ii) If $\delta_j > 0$, (2) makes $\{I_{j'} \cap E_j\}$, weighted $s_{j'}/\delta_j$, a fractional cover of E_j; $S \mapsto h(S \mid V_{j-1})$ is monotone submodular, hence fractionally subadditive, so $\sum_{j'} s_{j'} h(I_{j'} \cap E_j \mid V_{j-1}) \geq \delta_j g_j$. (iii) As $I_j \subseteq V_{j-1}$, $w_j h(E_j \mid I_j) \geq (1 - \delta_j) g_j$. Summing gives at least $\sum_j g_j = h(V)$. □

Only monotonicity and submodularity are used, so the argument holds over the whole polymatroid cone. The redundant coverage that lets per-step weights fall below 1—the source of every cyclic improvement, and the reason the elementary per-step argument ($w_j \geq 1$) does not apply—enters exactly as the fractional cover in (ii). The theorem assumes exact statistics of the database being bounded; the sketched ℓ_2 of Section VI-C is outside it.

a) Independent checks.: Two checks that do not rely on the proof agree with it. Over 207,000 databases—every database over a two-value domain for the triangle and 4-cycle, plus 15,000 random ones across six shapes at block sizes 1–3—the estimate never fell below the exact count (minimum ratio 1.0). And because the remaining inequality is linear in h and affine in w, an LP over the polymatroid cone decides it for every vertex of the feasible weight region: across 6,197 hyperpath structures on eleven shapes with $|V| \leq 6$ and block sizes 1–3 (1.73M vertices; every hyperpath on 19 of the 33 (shape, block size) pairs) the minimum slack is exactly 0—tight, never negative. Both ship as regression gates.

Theorem 5 (Subsumption)**.** *LpH-CEG subsumes:*

1) *Max-degree block bounds [5]: setting $p = \infty$, $w_j = 1$ recovers the Composition Lemma.*
2) *Binary ℓ_p-cover bounds: setting $|F_j| = 1$ recovers the fractional ℓ_p-cover over single relations—a polynomial-size relaxation of LpBound's [4] entropic LP (Section IV-A), not necessarily its entropic optimum.*
3) *AGM bound [1]: setting $p = 1$ on every step makes each attribute's coverage 1 per covering step, so (2) becomes the fractional edge-cover constraint and the objective becomes $\prod_j |R_j|^{w_j}$.*

A. Relationship to LpBound

LpBound [4] also applies to cyclic queries and also solves a linear program over ℓ_p-norm degree statistics, so we must delineate precisely what the block-cover LP adds.

a) Two different LPs.: LpBound's bound is the optimum of an *entropic* LP: it maximizes $h(V)$ over the Shannon cone—monotonicity and submodularity inequalities on all $2^{|V|}$ entropic terms—subject to the ℓ_p-norm statistics constraints, and is provably the tightest bound those statistics admit [4], [9]. Our LP (1)–(2) instead optimizes $O(m\,|\mathcal{P}|)$ weights over a *fixed hyperpath*, with one fractional-cover constraint per attribute. On identical statistics the entropic optimum is the tighter of the two by construction, and we do not claim otherwise: at bs=1 the cover bound is 1.25–1.43× looser on synthetic power-law cycles (Figure 2), 1.9–3.5× on six hand-picked real Hetionet cycles (`baseline_comparison.csv`), and up to 8.8× on one relation repeated along a long chain, where the entropic cone exploits conditional structure the fixed hyperpath treats independently.

b) On a fixed hyperpath, reweighting is exhausted.: Could a better weighting of the hyperpath's statistics close that gap? No. Since the remaining inequality is affine in w, one can minimize over a relaxation of the valid-weight set, test the minimizer with the polymatroid LP, and add the violating h as a cut. Run to convergence on triangle, cycle-4/5/6 and K_4, this returns *exactly* the cover bound (cycle-4: 15.335 in $\log_2$, against 14.858 entropic). On a fixed hyperpath the cover LP is already optimal over every valid weight vector; the residual gap comes from *statistics* the hyperpath leaves unused, which the entropic program exploits and Section VI-C extends.

The gap is predictable: over 198 configurations four structural features give a held-out R^2 of 0.87, and higher skew shrinks it (gap → 1 by α=3; ρ=−0.74).

c) Efficiency, stated correctly.: The Shannon-cone *formulation* of that optimum has $2^{|V|}$ variables and $O(2^{|V|}|V|^2)$ constraints and degrades accordingly: 111 ms at $|V|$=8, 43 s at $|V|$=12, ∼1 h at $|V|$=14, unsolvable beyond (Figure 2; the same curve on real Hetionet degree sequences: 46 s and 39 min). But that is not what LpBound solves. Its LPflow [4, Thm. 4.5] rewrites every simple ℓ_p constraint (one conditioning attribute) as a flow bundle after Im et al. [10] and is polynomial; our reimplementation returns the entropic bound to 10^{-6} on every instance we compared (12 families, the sweep, the 240 long-suite queries) in 3–26 ms at k=3–16 (Figure 2). So at bs=1 the cover LP buys *no* asymptotic advantage over LpBound, and we withdraw that reading: as a bs=1 bound the cover LP is dominated—looser, and both solve in milliseconds. The fixed hyperpath buys the per-step decomposition (item 1) and the framework the statistics vocabulary (item 2), where inference becomes non-trivial again (Section VI-E).

d) What the block-cover LP adds.: Not size or speed. What LpBound's single global program does not provide is structural:

1) **A per-step residual bound.** Because the LP is attached to a hyperpath, it yields a bound *per extension step*, and hence a residual bound for any variable prefix. This is exactly the signal our WCOJ variable ordering consumes (Section V); LpBound's single global optimum exposes no such per-step decomposition.
2) **Block-level statistics.** With multi-way blocks ($|F_j| > 1$) the LP draws on ℓ_p-norms of *joined* block relations—statistics outside LpBound's single-relation vocabulary.

So at bs=1 LpH-CEG trades a little tightness for a decomposable bound; item 2 is the axis the paper is about (Section VI-C).

Theorem 6 (Cyclic Improvement)**.** *Consider a cycle query C_k ($k \geq 3$) evaluated by binary extensions in cycle order, where relation R_j has cardinality $N_j = |R_j|$, maximum degree M_j, and degree-sequence ℓ_p-norms $\mu_p(R_j)$. Then:*

1) ***(Standard bound.)** The max-degree bound is*

$$B_{\max} = N_1 \prod_{j=2}^{k-1} M_j,$$

where the first extension has empty interface and contributes the cardinality N_1, the $k-2$ middle extensions each contribute a max-degree M_j, and the cycle-closing extension (interface of size two, empty extension set) contributes a factor of 1.

2) ***(LP bound.)** For every $p \geq 1$ the fractional block-cover LP (1)–(2) is feasible with weight $p/(p{+}1)$ and exponent p on every middle extension, together with the split $(\frac{p}{p+1}, \frac{1}{p+1})$ of unit weight across the first and closing (ℓ_1) extensions. This yields*

$$B_{\mathrm{LP}} \leq \min_{p \geq 1} \; N_1^{\frac{p}{p+1}} N_k^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \mu_p(R_j)^{\frac{p}{p+1}},$$

e.g. $N_1^{2/3} N_k^{1/3} \prod_j \mu_2(R_j)^{2/3}$ at p=2. (The exponent must be common, or non-increasing along the cycle: the cover constraint of the attribute between edges j and $j{+}1$ reads $\frac{p_j}{p_j+1} + \frac{1}{p_{j+1}+1} \geq 1$.)

3) ***(Improvement.)** Therefore, for every $p \geq 1$,*

$$\frac{B_{\max}}{B_{\mathrm{LP}}} \;\geq\; \Big(\frac{N_1}{N_k}\Big)^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \frac{M_j}{\mu_p(R_j)^{\,p/(p+1)}}.$$

At p=2 each middle factor exceeds 1 when the relation is skewed ($\mu_2(R_j) < M_j^{3/2}$); as $p \to \infty$ every factor tends to 1. The boundary factor is a constant independent of

k; for a uniform *cycle ($N_j \equiv N$) it equals* 1*, and in the general case it can be made* ≥ 1 *by entering the cycle at the largest relation ($N_1 \geq N_k$). The improvement is therefore dominated by the product of $k-2$ per-edge factors and grows geometrically in the number of middle edges $k-2$. Since LpH-CEG returns* $\min(B_{\max}, B_{\mathrm{LP}})$*, the realized ratio is always* ≥ 1.

Proof sketch. The standard bound (1) follows directly from the cycle hyperpath structure: the first extension has empty interface, so its degree sequence is the single value N_1 and $\mu_\infty = N_1$; each middle extension adds one attribute over a size-one interface with $\mu_\infty = M_j$; the closing extension has both endpoints already in the state, so its conditional degrees are all 1 and $\mu_\infty = 1$. For (2), each attribute is covered as an extension in one step (rate 1) and as an interface in the next (rate $1/p$); with a common p the local constraint $w + w/p \geq 1$ is tight at $w = p/(p+1)$. Attribute a_2 then reads $w_1 + \frac{1}{p+1} \geq 1$, a_k reads $\frac{p}{p+1} + w_k \geq 1$ and a_1 reads $w_1 + w_k \geq 1$, all tight at $w_1 = \frac{p}{p+1}$, $w_k = \frac{1}{p+1}$. The closing step has $\mu_1 = N_k$, so the first/closing contributions are $N_1^{p/(p+1)} N_k^{1/(p+1)}$ against N_1 in $B_{\max}$. The LP optimum is at most every such point, and may mix p further. □

Corollary 7 (Scalability with cycle size)**.** *For cycle-k queries on power-law graphs with fixed parameters, the LpH-CEG improvement ratio grows exponentially with k. Our experiments confirm (10-seed mean):* $8.4\times$ *(k=4),* $69\times$ *(k=6),* $614\times$ *(k=8).*

B. Tightness and Limitations

Theorem 4 establishes that LpH-CEG bounds are *valid* (never underestimate), but not necessarily *tight* (matching a worst-case instance). Tightness depends on the degree sequence structure:

Proposition 8 (Per-step tightness)**.** *Each step's ℓ_p inequality is tight when its interface is simple ($|I_j| \leq 1$) [9]. Under binary extensions most interfaces are simple, the exceptions being the first step ($I_1 = \emptyset$) and any closing step, whose interface carries the two endpoints of the closing edge.*

Per-step tightness does not make the *composition* tight: the weighted product can be loose even where every factor is exact, and we measure it to be 1.25–8.8$\times$ looser than the entropic optimum at bs=1 (Section IV-A).

This creates a principled trade-off: binary extensions keep simple interfaces; multi-way blocks capture block-level correlation at the cost of non-simple ones. Results use bs=1 unless marked bs=2.

C. Delta over the Block Framework

We inherit the block-extension DAG and hyperpath machinery from H-CEG [5], which uses only ℓ_∞ weights with no optimization layer. What this paper adds is the ℓ_p fractional cover LP that replaces those fixed scalars, the separation of the statistics and inference axes (Section VI-C), the flow-LP inference over block statistics (Section VI-E), and the per-step residual signal; none exists in H-CEG. Theorem 6 additionally supplies a structural test neither predecessor provides, predicting the magnitude from a product of $k-2$ per-edge factors $M_j/\mu_p(R_j)^{p/(p+1)}$ at a common p.

Algorithm 2 LpH-CEG Variable Ordering for WCOJ

Require: Query H, relation data D
Ensure: Variable ordering $\sigma = (v_1, \ldots, v_n)$
$\sigma \leftarrow []$, chosen $\leftarrow \emptyset$
for $i = 1$ **to** n **do**
  for each $v \in V \setminus$ chosen **do**
    $b_v \leftarrow$ LpH-CEG-Bound$(H|_{V \setminus (\text{chosen} \cup \{v\})}, D)$
  end for
  $v^* \leftarrow \arg\min_v b_v$
  Append v^* to σ; add v^* to chosen
end for

V. WCOJ Variable Ordering

LpH-CEG bounds naturally guide WCOJ variable ordering. At each step of the greedy ordering, we select the variable whose removal minimizes the LP-optimized residual bound (Algorithm 2):

On the Hetionet cycle-4 query, the LP-guided ordering (gene1→gene2→compound→disease) differs from the standard max-degree ordering (compound→disease→gene2→gene1) and reduces GenericJoin iterations from 2,765,886 to 1,209,687—a 2.3$\times$ reduction (a deterministic, machine-independent metric) resulting in 2.6$\times$ faster wall-clock execution in our simulator.

a) Does it generalize? An honest negative.: It does not, and we report this plainly. On a broad synthetic suite (66 queries across seven shapes, two sizes, two skews, and three seeds; joins too large to enumerate are censored), the LP-residual ordering is *not* a general improvement over the max-degree (H-CEG) residual ordering: the median iteration ratio is 0.99 (-1.3%), it wins on 20 queries but loses on 43 (Wilcoxon p=0.003, Cliff's δ=-0.35), and its 90th-percentile regression is 1.8$\times$. The Hetionet win reflects a regime (heterogeneous relations with asymmetric skew), not a universal effect, so we treat LP-guided ordering as a targeted heuristic, not a systems contribution (see also Section VI-G).

VI. Experimental Evaluation

A. Setup

a) Datasets.: We evaluate on three real-world datasets: **Hetionet** [11] (biomedical knowledge graph, 45K nodes, 2M edges, heterogeneous); **Epinions** (social network, 75K nodes, 509K edges); **IMDb**, JOB's snapshot [12]: the 200 most-voted feature films for a cycle suite, and the JOB-medium subset. We also use synthetic power-law graphs (100–5,000 nodes, 400–20,000 edges). All synthetic experiments report mean $\pm$ std over 10 random seeds.

b) Queries.: Synthetic cycle-k for $k \in \{3, \ldots, 16\}$ plus diamond and K_4 families; on real data, three registered suites (Section VI-D) plus a handful of hand-picked cycles.

TABLE I

UNIFIED COMPARISON OF GUARANTEED UPPER BOUNDS (SYNTHETIC POWER-LAW CYCLES, 100V/400E). "COVER" IS OUR bs=1 LP; "ENTROPIC" IS THE $2^{|V|}$ LPBOUND. LOWER IS TIGHTER. "/CEG" AND "/ENT" ARE COVER'S RATIO TO CEG_M AND TO THE ENTROPIC OPTIMUM.

Query	AGM	CEG_M	Cover	Entropic	/CEG	/Ent
Cycle-3	8.0K	15.6K	6.8K	5.0K	2.3×	1.36×
Cycle-4	160K	655K	116K	87K	5.7×	1.32×
Cycle-5	3.2M	25.5M	1.97M	1.45M	13.0×	1.36×
Cycle-6	64M	1.12B	31.5M	25.2M	35.7×	1.25×

TABLE II

LPH-CEG VS. CEG_M ON SYNTHETIC CYCLE QUERIES (bs=1, $|V|$=200, $|E|$=800, MEAN ± STD OVER 10 SEEDS). THE RATIO IS A *bound-tightening* MEASURE (CEG_M/LPH-CEG); THE q-ERROR COLUMN (ACCURACY VS. THE TRUE COUNT) IS SHOWN ONLY WHERE THE EXACT COUNT IS FEASIBLE TO MATERIALIZE, AND IS "—" FOR $k \geq 6$ WHERE THE TRUE CARDINALITY IS INTRACTABLE TO COMPUTE.

Query	CEG_M	LpH-CEG	Ratio	q-error
Cycle-3	53K±2K	19K±0.3K	2.75±0.13×	10.8±0.2×
Cycle-4	3.8M±253K	448K±44K	8.43±0.78×	18.3±1.8×
Cycle-5	270M±18M	11.3M±308K	23.8±1.3×	39.0±1.1×
Cycle-6	18.8B±2.3B	272M±16M	68.9±6.4×	—
Cycle-7	1.4T±167B	6.5B±329M	211±20×	—
Cycle-8	97T±14T	158B±10B	**614±71**×	—

c) Baselines.: AGM (fractional edge cover, $\prod |R_i|^{w_i}$); CEG_M (binary max-degree shortest path, bs=1; equivalent to PANDA/MOLP on binary queries [2], [3]); LpH-CEG (LP-optimized ℓ_p-norms, $\mathcal{P} = \{1, 2, \infty\}$). The "LpH-CEG" column throughout denotes our *cover*-LP at bs=1 (Section IV-A); it is *not* the entropic LpBound of Zhang et al. [4], against which we compare separately in Section IV-A (our cover bound is 1.3× looser on synthetic cycles, up to ∼9× on homogeneous real chains). Learned and summary estimators (FactorJoin, GNCE, COLOR [13], [14]) produce *point* estimates without guaranteed upper bounds; CorrBound, which enriches the entropic program with cross-relation statistics, is compared through a proxy in Section VI-C.

d) Unified bound comparison.: Table I places all four guaranteed-bound methods on one suite. The ordering is as predicted: $\text{CEG}_M \geq$ Cover-LP $\geq$ Entropic, with AGM incomparable (it ignores degrees). Our cover-LP is 2.3–35.7× tighter than CEG_M on synthetic cycles, is also consistently tighter than AGM (e.g. 2.0× on cycle-6), and stays within 1.25–1.36× of the entropic optimum—the best any ℓ_p-statistics bound can achieve.

B. Main Results: Cyclic Query Bounds

Table II shows the core result: LpH-CEG achieves 2.8× to 614× tighter bounds than CEG_M on cyclic queries, with tight standard deviations (relative std $< 12\%$) confirming stability across random graph instances. The improvement *scales with cycle size*: from 8.4× on cycle-4 to 69× on cycle-6 to 614× on cycle-8. It also *scales with data size*: cycle-4 improves from 5.9× (100V) to 34× (2KV), with the dependence on skew characterized in Section VI-H.

TABLE III

STATISTICS × INFERENCE, 40 VERTICES / 120 EDGES, SEED 101 (12 FAMILY–SEED PAIRS BEHAVE ALIKE; SEE TEXT). BOTH PROGRAMS SEE THE SAME CATALOG: EVERY SINGLE RELATION, OR EVERY $bs \leq 2$ BLOCK AND INTERFACE. EVERY ENTRY IS A VALID UPPER BOUND ON THE EXACT COUNT.

	cover LP		polymatroid optimum	
Query	single	block	single	block
Triangle	1,083	668	806	**368**
Cycle-4	9,871	4,716	7,436	**1,355**
Diamond	9,769	4,560	7,436	**1,355**

C. Statistics Versus Inference

The two axes can be separated directly: hold the program fixed and change the statistics, then hold the statistics fixed and change the program. Table III is that 2×2, over four families and three seeds, with the exact join size computed so every bound is checked rather than assumed.

Reading the table by column pairs gives the central observation. At fixed program, the statistics axis (single → block) improves the bound by 1.5–2.3× for the cover LP and 1.7–5.5× for the optimum; at fixed statistics, the inference axis (cover → optimum) improves it by 1.15–1.5× on single-relation statistics and 1.5–3.8× on block ones, over all 12 pairs. The axes interact: richer statistics give the program more to exploit. The statistics axis has room because a block statistic $\mu_p(F \mid I)$ with $|F| \geq 2$ describes a multi-relation join, which no single-relation degree *sequence* determines (most of it follows from single-relation degree *tables*; see below).

Two scoping points keep this from being overstated, both measured on the headline cycle protocol ($|V|$=200, $|E|$=800, 10 seeds) so the axes are in the same units:

Axis	cycle-4	cycle-6	cycle-8
inference: max-deg → cover LP	8.8×	72.7×	654×
inference: cover LP → entropic		1.25–1.43×	
statistics: bs=1 → bs=2	1.21×	1.87×	2.70×

First, the claim concerns headroom *remaining once an LP is in place*. The first inference step—from a max-degree product to the cover LP—dwarfs both of the others and is the contribution of the LP itself. Second, the separation between the two remaining axes is regime-dependent: on these homogeneous synthetic cycles they are comparable (1.2–2.7× against 1.25–1.43×), and it is heterogeneous real data that separates them, where the same comparison is 56× against 2.6×.

The diagonal comparison is the practical one. The *cheap* LP on rich statistics (4,716) beats the *optimal* LP on poor statistics (7,436)—on 12/12 family–seed pairs, and on all 109 cycle-3/4 queries of the pre-registered Hetionet suite (rows capped at 3K so the pair joins stay small), where the median ratio to the entropic bound is 0.057× and the 20 tightest bounds were checked against exact join sizes with no violation. On those 20 the true counts are only 0–70 against bs=2 bounds of 291–827: the ordering moved, not the distance to the truth—Table IV is

the uncapped test. We stress the direction of this claim: it does *not* say the cover LP is a better program. At equal statistics the optimum wins on $12/12$ pairs, as it must. It says that the choice of statistics comes first, and it identifies the strongest cell—the optimum on block statistics, i.e. LpBound run on materialized pair-join views, which no published evaluation does; Section VI-E computes it in polynomial time.

Table IV carries the comparison to the sizes the introduction motivates, $|V|$=6–10, on two registered suites (Section VI-D). On Hetionet the two axes diverge with $|V|$: the inference gap stays at 2.5–3.6× (max 16×), while bs=2 statistics are 80×/630×/2,500× tighter than the entropic bound at the median, on $120/120$ queries, with none of the 345 validity checks failing. The exact counts are 10^4–10^9, so this is not a small-count artifact; the entropic bound is $10^8\times$ the truth at $|V|$=10 and the flow LP on bs=2 brings that to $5.6{\cdot}10^3\times$. Two more rows locate *which* statistic carries the gain. Keeping only the *sizes* of the same pair joins, $|R \bowtie S| = \sum_b \deg_R(b)\deg_S(b)$—the two-relation statistic CorrBound [15] adds to the entropic program, here added exactly (the constraints stay simple, so the flow LP is the polymatroid optimum) though without CorrBound's sketches or multi-relation terms—already beats the entropic bound by a median 15/27/99× (AGM over the same sizes: 7–15×). Under the same program (the flow LP, within 1.73× of the optimum, so these are lower bounds) the interface-conditioned block statistics add a further 19/80/154× on $120/120$ queries (never below the exact count). Join sizes matter; conditioning on the interface matters more.

On Hetionet the price is on the other side of the ledger: bs=2 statistics take 25–57 s at the median (up to 366 s; 33 queries exceeded a 180 s budget and are flagged, not dropped), against under 1 s for bs=1 and 0.5–2.2 s for the entropic LP at these sizes. Those seconds rebuild per query what a system materializes once. The last row measures that: a workload-level catalog keyed by underlying table, orientation and interface holds 676 entries from 91 block builds, costs 527 s in total against the 11,765 s the per-query script spent, and returns the identical bound on $120/120$ queries. It saturates by the 68th query, after which a bs=2 estimate costs 7–10 ms, 100–220× below the entropic LP at $|V|$=8–10 and below it on $60/60$ of the second half. The 45× collection overhead is real, but once per database.

IMDb. The second suite is a different kind of graph—a star schema in which every relation joins movies to people, companies or keywords—and reproduces the ordering with smaller margins. The inference gap is 3.8–4.3×; bs=2 statistics are 4.0/13/43× tighter than the entropic bound under the cover LP and 17/53/177× under the flow LP, on $120/120$ queries, so at $|V|$=6 the cover-LP comparison is level with the inference gap. Under the flow LP the bound moves from 74–9,200× the truth to 4–60×; none of the 101 checked queries is violated (exact counts $3{\cdot}10^5$–$9{\cdot}10^7$ at the median; 19 of the 80 at $|V|$=8–10 timed out). Two differences are informative. Pair-join sizes barely help (median 1.00×; they help at all on $24/120$, by at most 1.35×): half the pairs share an entity and so join one relation with itself, with size $\sum_x \deg(x)^2$, that relation's ℓ_2^2, which LpBound already has; the movie-sharing pairs are genuine inner products but rarely bind. The interface-conditioned statistics still add 17–171×. And at 200 films the program gap does not grow with the statistics (3.3–4.2× at bs=2 against 3.8–4.3× at bs=1); it does at scale, below. (IMDb times were taken under 10 concurrent workers and IMDb has no catalog run; the 45× overhead and the 10 ms cataloged estimate are Hetionet's.) At five times the films (1,000; the 40 queries at $|V|$=6, bounds only) every tightness ratio grows: the inference gap from 3.8 to 4.8×, block statistics from 4.0 to 7.0× under the cover LP and from 17 to 48× under the flow LP, $40/40$ below entropic—and the program gap from 4.2 to 6.8×, now above the inference gap, so on IMDb the reopening appears with scale. Collecting the statistics grows faster than the data: 4.2× the rows cost 48 s per query at the median (from 4.4 s), up to 25 min and 16 GiB, measured with 10 workers sharing the machine.

a) Which statistic needs the join.: Of the eight interfaces of a block of binary relations $R(a,b), S(b,c)$, seven (five non-trivial) are functions of single-relation degree tables and cost one pass over each relation: the degree of a is $\sum_{b\in N_R(a)} \deg_S(b)$ and that of b is $\deg_R(b)\deg_S(b)$; only $\{a,c\}$, the common-neighbour count $|N_R(a)\cap N_S(c)|$, needs the join. We compute the seven without it (identical to the materialized statistics on $120/120$ Hetionet queries; 0.49 s against 4.6 s per block). Added first, they close most of the log-gap to the full bound but not the last factor: a median 2.3× (max 18.7×) above the full bs=2 bound on the Hetionet long suite, 1.5–2.9× on the families, 1.9–57× on the sweep. What closes it is one number per block, the ℓ_2 norm of the common-neighbour sequence, $\|A_R^\top A_S\|_F$ for the $b\times a$ and $b\times c$ incidence matrices. Added to bs=1 alone it already reaches a median 1.03× of the full bound ($53/120$ exact), so the attribution is order-dependent; added to the join-free seven it returns the full bound on $12/12$ families, $10/10$ sweep points, $85/120$ Hetionet and $116/120$ IMDb queries (Hetionet median 1.00×, max 1.67×), while its ℓ_∞ does not (1.70×) and its ℓ_1—the join size, CorrBound's inner product [15]—is already join-free. The one-pass upper bound on that norm from the triangle inequality recovers nothing on Hetionet (2.2×); a 64-probe Hutchinson sketch lands within 0.91–1.78× of the full bound (median 1.04× on Hetionet, above the truth on all 115 checked) in 0.32 s per block—an estimate, not a bound, so a cost floor rather than a method. The expensive statistic is the necessary one.

The same reading resolves an apparent contradiction: on cycle-3/4/5 the bs=1 LP tightens CEG_M by 2.9/8.5/25.9×, max-degree *blocks* with no LP by *more* (4.0/11/45×), and the cover LP on top of bs=2 changes nothing—not because the program cannot matter there (on the cycle-3/4/5 sweep instances the flow LP gains 2.2–7.1× over it on the same statistics), but because one hyperpath uses too few of them.

Over k = 3–10 (10-seed means) the bs=1 improvement doubles roughly every 1.3 steps of k, as Corollary 7 predicts.

TABLE IV

TWO REGISTERED REAL SUITES (IMDB AFTER A DISCLOSED PILOT), 40 CYCLES PER LENGTH ADMITTED IN SHA-256 ORDER OF THE CANONICAL RELATION SEQUENCE. HETIONET: METAGRAPH CYCLES OVER DISTINCT RELATIONS, 20K ROWS PER RELATION. IMDB (JOB SNAPSHOT): THE 200 MOST-VOTED FEATURE FILMS, NINE CAST ROLES, TWO COMPANY TYPES AND KEYWORDS, ONE RELATION EACH. MEDIANS OVER QUERIES; "FLOW" IS THE FLOW LP OVER THE $bs{\leq}2$ CATALOG. EXACT COUNTS BY DUCKDB (60 S BUDGET); QUERIES THAT TIME OUT ARE UNCHECKED, NOT DROPPED. IMDB TIMES WERE TAKEN UNDER 10 CONCURRENT WORKERS. THE CATALOGED ROW AMORTIZES BLOCK BUILDS OVER THE WORKLOAD; THE $|V|{=}6$ QUERIES, RUN FIRST, PAY MOST OF THEM.

	Hetionet			IMDb		
$\|V\|$	6	8	10	6	8	10
max-degree / cover $bs{=}1$	$1.8\times$	$5.1\times$	$8.9\times$	$1.8\times$	$1.9\times$	$3.2\times$
cover $bs{=}1$ / entropic	$2.53\times$	$3.24\times$	$3.61\times$	$3.75\times$	$4.33\times$	$4.14\times$
cover $bs{=}2$ / entropic	$0.012\times$	$0.0016\times$	$0.00039\times$	$0.25\times$	$0.075\times$	$0.023\times$
flow $bs{=}2$ / entropic	$0.0035\times$	$0.00038\times$	$0.000063\times$	$0.060\times$	$0.019\times$	$0.0056\times$
$bs{=}2$ below entropic	$40/40$	$40/40$	$40/40$	$40/40$	$40/40$	$40/40$
entropic+pair sizes / entropic	$0.067\times$	$0.037\times$	$0.010\times$	$1.00\times$	$1.00\times$	$1.00\times$
flow $bs{=}2$ / entropic+sizes	$0.054\times$	$0.012\times$	$0.0065\times$	$0.060\times$	$0.019\times$	$0.0059\times$
cover / flow, both $bs{=}2$	$3.2\times$	$4.2\times$	$5.1\times$	$4.2\times$	$4.0\times$	$3.3\times$
flow $=$ polymatroid	39	35	30	28	27	30
entropic / exact	$1.5{\cdot}10^4$	$1.1{\cdot}10^6$	$1.0{\cdot}10^8$	74	970	$9.2{\cdot}10^3$
flow $bs{=}2$ / exact	81	406	$5.6{\cdot}10^3$	4.2	16	60
checked (invalid)	40 (0)	40 (0)	35 (0)	40 (0)	33 (0)	28 (0)
seconds, $bs{=}1$/ent/$bs{=}2$	0.6/0.5/25	0.8/0.8/57	0.8/2.2/33	0.3/0.3/4.4	0.6/0.8/14	0.9/3.3/22
$bs{=}2$, cataloged	0.56	0.007	0.010	—	—	—

D. Real Data Results

At $bs{=}1$ on real data the LP fires only where cyclic structure meets cross-relation degree asymmetry (Section VI-I). The heterogeneous Hetionet cycle-4—four relation types with asymmetric degrees—improves $7.59\times$ (53M→7.0M, relations capped at 30K edges); the low-skew Hetionet triangle and the replicated Epinions cycles on 2–5K-edge subsets yield exactly $1.00\times$ (at 20K edges the Epinions cycle-4 improves $2.06\times$, Section VI-D). Over six closed metapaths of length 3–5 built from Hetionet's thirteen edge types, LpH-CEG fires *selectively and never spuriously*: it tightens C-G-G ($1.95\times$), C-D-G-G and C-D-A-G-G ($7.59\times$)—exactly those whose hyperpath places the skewed gene–gene relation in a *middle* position—and returns $1.00\times$ on the rest. The effect is *structural*: D-G-G-A holds the same high-skew relation as C-D-G-G yet yields $1.00\times$ because it sits on the closing edge, as Theorem 6 predicts.

Beyond cycles. On a diamond, a 4-clique and a chordal 4-cycle over the same edges LpH-CEG improves 1.9–$4.2\times$ over max-degree (largest on the 4-clique): it fires wherever a skewed relation sits in a middle position, not only on cycles.

A pre-registered suite. A handful of hand-chosen queries cannot support a distributional claim, so we fixed a suite *before* measuring: every type-consistent metagraph cycle of length 3–5 over distinct Hetionet relations (370 queries, admitted by rule), plus seven Epinions shapes. The registry is hash-pinned and the evaluation refuses to run against an edited copy, so selection cannot have followed the outcome. All 377 were evaluated and *none* censored. Over the 370 Hetionet queries the cover/entropic gap has median $1.99\times$, P90 $3.55\times$, maximum $6.29\times$ (215 distinct values), meeting the median $\leq 3\times$, P90 $\leq 8\times$ bar registered in advance. Improvement over max-degree has median $1.47\times$ and reaches $13.1\times$, yet $112/370$ queries (30%) improve by exactly $1.00\times$—the selective behavior above, now over a population rather than a handful. The Epinions half is degenerate and we report it so: one homogeneous relation self-joined makes all seven shapes yield an identical cover/entropic *gap* of $3.28\times$—one distinct observation rather than seven. A second registry extends the rule to lengths 6, 8 and 10, where the 777/4,380/12,960 admissible cycles are too many to solve entropically, so the first 40 per length in SHA-256 order of the canonical relation sequence are admitted—an order blind to data and bounds—giving the 120 Hetionet queries of Table IV. A third registry, fixed after a disclosed pilot, uses IMDb from the Join Order Benchmark [12]: the 200 most-voted feature films (197–37K rows per relation), with one relation for each of nine of the twelve cast roles (costume and production designers and guests omitted), for distributors, for production companies, and for keywords. Each relation joins movies to an entity type of its own, so a person reached as an actor and as a director are different variables: cross-role joins are excluded by design. The schema is then a star—a cycle alternates movies with entities, each entity reached by one relation on both sides, and relations repeat (JOB joins `movie_companies` and `title` twice); of the 364/3,081/25,752 such cycles at length 6/8/10 the first 40 in hash order are admitted. The pilot used the Hetionet rule—one person type, distinct relations—on the first 3,500 feature films of the year 2005; $82/120$ of its joins were empty, so validity could not be tested, and scope and typing were both changed before registering. The pilot favored block statistics more (median cover $bs{=}2$/entropic $9.5{\cdot}10^{-6}$ against 0.077 here); it, and a run at five times the films (Section VI-C), are in the artifact.

E. Scalability

On power-law cycles ($|V|$=100, $|E|$=400, median of 7 runs) the LP adds under 4.7 ms to the max-degree bound (1.17–1.34× overhead, 8–24 ms in total for k=3–10) while the improvement grows exponentially with k, reaching 971× at cycle-10 (`lp_scalability.csv`).

a) Polynomial inference on block statistics.: The Shannon-cone formulation of the entropic bound has $2^{|V|}$ variables and, on the power-law cycles of Figure 2, rises from 111 ms at k=8 to 43 s at k=12 and 3,679 s at k=14 (real Hetionet sequences: 46 s and 39 min). LpBound does not solve that program. Its LPflow [4, Thm. 4.5] follows Im et al. [10]: each *simple* constraint ($|X| \leq 1$) becomes a capacitated edge $X \to Y$ in a flow network over attribute sets (an ℓ_p constraint adds $\emptyset \to X$ at capacity δ/p), and the bound is the cheapest capacity assignment that routes one unit of flow to every attribute—a polynomial LP. Our reimplementation returns the entropic bound to 10^{-6} on all 12 instances of Table III and at every $k \leq 14$ here, in 3–26 ms for k=3–16; the cover LP is 1.25–1.43× looser than a program that is itself polynomial; both solve in milliseconds.

The question that remains open is inference over *block* statistics, where a bs=2 block over three attributes contributes constraints conditioned on two of them ($|X|$=2)—the non-simple case, which Im et al. prove hard at $|X| \leq 2$, $|Y| \leq 3$. Their §7 relaxation handles such a constraint along a fixed attribute order π by restricting it to the attributes of Y that follow all of X; we carry the ℓ_p term through unchanged. Validity for every order follows from their argument; we checked it on 40 random connected shapes ($|V| \leq 5$, all 2,046 order–instance pairs): never below the relaxed polymatroid bound or the exact join. Two facts make the relaxation usable. The $|X|$=2 constraints carry most of the bs=2 gain (dropping them loosens the polymatroid bound 1.5–2.9×), and an order keeps one exactly when both conditioning attributes precede its target—so on a cycle an *alternating* order (one color class, then the other) keeps every constraint centered on the second class; no single order keeps them all, so we solve the flow LP over the $4|V|$ walk and alternating orders and keep the best bound. Over the full bs=2 catalog (every block, every interface) it equals the $2^{|V|}$ polymatroid bound on 12/12 family–seed pairs and on every synthetic even cycle up to k=10 (the largest the $2^{|V|}$ program solves, 0.7 s) and on the triangle, and loses a constant 1.56–1.59× on cycle-5/7, where one wedge must remain lost; 34/40 of the random shapes are exact (max gap 1.28×). Cost is 4–73 ms per order and 0.03–4.4 s in total at k=3–16 (Figure 2). What it *finds* matters more than what it costs: the cover LP, confined to one hyperpath, leaves most of the block statistics unused, and the flow LP over the same catalog is 1.5–3.8× tighter on the small families and 2.2× at k=3 growing to 288× at k=16 on the sweep (below the bs=1 optimum by 2–800× where the cover LP manages 1–6×; Figure 1). On the 120-query Hetionet long suite ($|V|$=6/8/10) the median gain over the cover LP on the same statistics is 3.2/4.2/5.1×, the flow LP

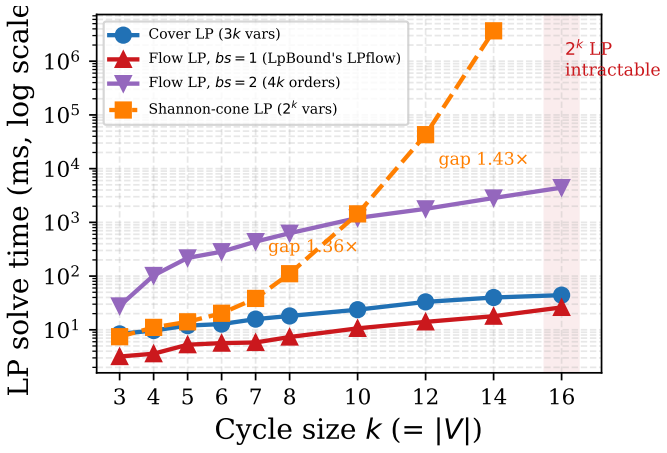


Fig. 2. LP solve time on cycle-k queries (log scale). The Shannon-cone formulation of the bs=1 entropic bound grows super-exponentially (43 s at k=12, 3,679 s at k=14), but LpBound's flow LP reaches the same bound (annotated gap to the cover LP) in 3–26 ms. On bs=2 block statistics the flow LP over $4k$ orders takes 0.03–4.4 s and equals the $2^{|V|}$ polymatroid bound at every even k of this sweep. Data: `scale_entropic.csv`, `flow_scale.csv`.

matches the $2^{|V|}$ polymatroid bound on 104/120 queries (max gap 1.73×), never falls below the exact count on the 115 with one, and takes 0.27–1.1 s (median) against 0.5–2.2 s for the $2^{|V|}$ program at bs=1.

b) Choice of p-values.: On cycle-6, using $\mathcal{P} = \{1, \infty\}$ (2 values) already achieves 17.6× improvement; adding ℓ_2 ($\mathcal{P} = \{1, 2, \infty\}$) reaches 35.7×; further values (ℓ_3, ℓ_5) yield no additional gain (Figure 3). We recommend $\mathcal{P} = \{1, 2, \infty\}$ as the default (cardinality ℓ_1, variance ℓ_2, worst case ℓ_∞); repeating the sweep on four families (cycle-4/6, K_4, chordal-5) confirms this is not cycle-specific—ℓ_2 helps in every family and p beyond $\{1, 2, \infty\}$ adds $< 2.4\%$.

F. WCOJ End-to-End Execution

We evaluate ordering quality in a controlled GenericJoin *simulator* (a faithful Python implementation that counts tuple-level iterations); the primary metric is the *iteration count*, which is deterministic and engine-independent, while wall-clock time is reported as a single-machine reference and varies ±10% across runs. To confirm the simulator computes the correct join, we cross-validate its output cardinalities against Kuzu—an independent WCOJ-capable graph engine—on cyclic patterns: the counts match exactly across all tested configurations. A full integration that *injects* our ordering into a production engine is left to future work; we found that neither DuckDB nor Kuzu exposes the per-variable ordering hook this would require, so it entails modifying an open-source WCOJ engine's planner.

Table V and Figure 4 demonstrate end-to-end impact. On the heterogeneous Hetionet cycle-4, LpH-CEG ordering (gene1→gene2→compound→disease) reduces iterations by 2.3× and simulator wall-clock time by 2.6× versus max-degree ordering (compound→disease→gene2→gene1). The LP selects an ordering that processes high-selectivity joins

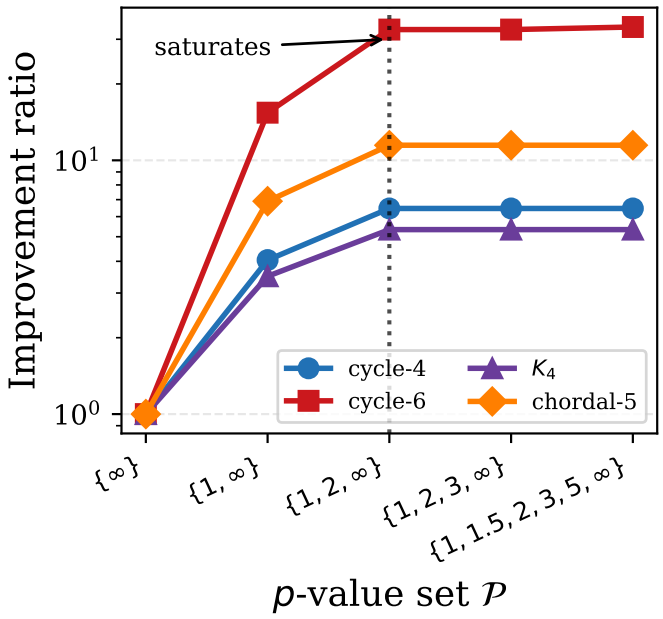


Fig. 3. Sensitivity to the number of p-values across four query families (log scale). Improvement saturates at $\mathcal{P} = \{1, 2, \infty\}$ (dotted line) in every family; additional norms add $<2.4\%$.

TABLE V
WCOJ GENERICJOIN EXECUTION WITH DIFFERENT VARIABLE ORDERINGS. LPH-CEG SELECTS ORDERINGS THAT REDUCE INTERMEDIATE RESULT SIZES.

Query	Strategy	Iterations	Time (ms)	Speedup
Hetionet C-4	Max-deg	2,765,886	1,455	1.0×
	LpH-CEG	1,209,687	560	**2.6**×
Cycle-4 synth.	Max-deg	258,068	1,063	1.0×
	LpH-CEG	260,463	1,059	1.0×
Triangle	Max-deg	38,757	72	1.0×
	LpH-CEG	38,757	73	1.0×

first, pruning intermediate results earlier. On homogeneous synthetic data and acyclic queries, both orderings perform identically, confirming that LP overhead does not harm performance when improvement is unavailable.

G. End-to-End: Robustness of Bound-Guided Ordering

The WCOJ result above guides a multi-way join. We now ask the more demanding systems question: can the cover bound guide *binary* join ordering inside a modern engine, and how does that compare to the engine's own optimizer? We load each query into DuckDB, compute a left-deep order by minimizing the classic C_{out} cost [12] (sum of the bound's estimated intermediate cardinalities), inject it by disabling DuckDB's join-order optimizer, and time real execution. As a reference we also run DuckDB's own sampling-based optimizer.

Two honest findings frame the result. First, a cardinality *upper bound* is a conservative cost model: it never underestimates, so ranking plans by it does not beat—and need not even match—an engine optimizer on average. Second, and structurally, on a single cycle every proper sub-query is acyclic, where the LP has no cyclic slack (Proposition 2), so the two bounds largely induce the same order; the bound changes the ranking only when an *intermediate* sub-query is itself cyclic (cliques, chordal cycles).

TABLE VI
END-TO-END ROBUSTNESS: SLOWDOWN OVER THE BEST EVALUATED PLAN (DUCKDB, FORCED LEFT-DEEP ORDER VS. THE ENGINE'S OWN OPTIMIZER). LOWER IS BETTER; THE *tail* ROW IS THE WORST CASE OVER ALL QUERIES.

Query	Bound-guided	DuckDB opt.
Cycle-4 (100V/400E)	1.10×	0.60×
Cycle-5 (100V/400E)	1.52×	0.14×
K_4 clique (120V/800E)	1.64×	18.1×
K_4 clique (120V/1500E)	3.32×	21.0×
K_4 clique (120V/2500E)	1.81×	7.2×
K_4 clique (200V/2000E)	1.57×	18.3×
Diamond / chordal-4	2.54×	0.39×
Chordal 5-cycle	2.11×	0.13×
Hetionet △ (real)	3.56×	1.11×
Worst case (tail)	**3.6**×	**21.0**×

What the pessimistic bound *does* buy is robustness against catastrophic plans. Table VI reports, per query, each strategy's slowdown over the best plan we found. DuckDB's optimizer is excellent on average—often faster than any forced left-deep order because it builds bushy plans—but on dense 4-cliques its sampling-based estimates badly underestimate the cyclic intermediate blow-up, and it runs 7–21× slower than optimal across every clique density we tried. Bound-guided ordering, which by construction cannot be lured into such a plan, stays within 3.6× of optimal on *every* query in the suite. The worst-case (tail) slowdown is 3.6× for bound-guided versus 21× for the engine—a 5.9× heavier tail for the sampling-based optimizer.

a) The tail, over a broader distribution.: Of six shapes at two sizes and three seeds, 24 queries survive (cycle-3/4, K_4, diamond; cycle-5 and chordal-5 censored entirely). Medians are close (1.03× vs. 0.93×) while the tails diverge: the bound-guided P95/max are 1.21×/1.31× against DuckDB's 5.82×/8.17×. Censoring uses the AGM bound, which depends only on relation sizes—censoring by our own bound would drop exactly the queries where it is loose. One caveat: the reference is the best *forced left-deep* order, so bound-guided is $\geq$1 by construction while DuckDB, free to go bushy, scores below 1 on median. This shows a bound-guided choice avoids a plan space's tail, not that it beats the engine.

b) Real data, block statistics.: The same protocol on the 40 IMDb cycles at $|V|$=6, with four estimators behind C_{out}, tests the thesis rather than the bs=1 bound. Block statistics move the choice toward the true-cardinality order: the flow LP on bs=2 picks the order exact counts would pick on 24/40 queries, and on the 33 whose forced orders all finish within 60 s its median slowdown over the best timed order (the strategies' picks plus 16 sampled, of the 96 connected left-deep orders) is 1.19×, against 1.41× for the bs=1 cover LP, 1.23×

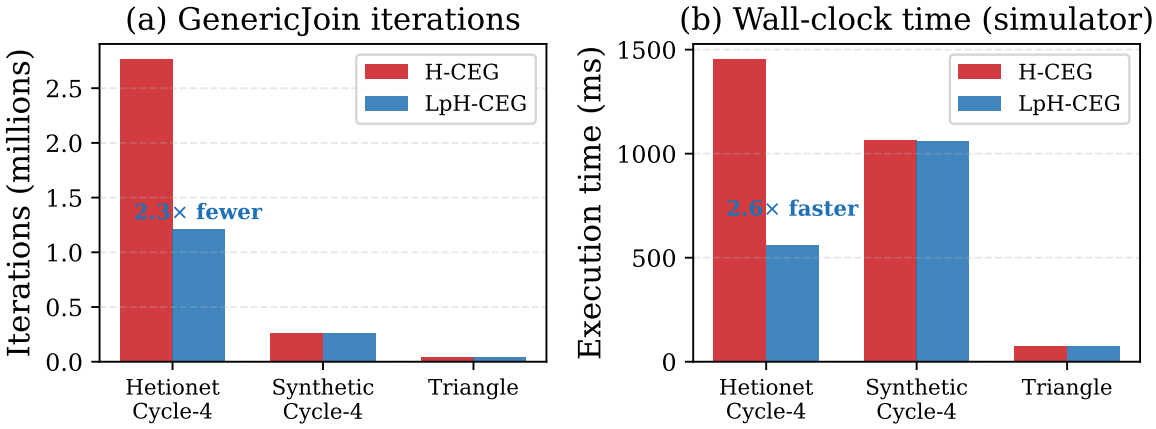


Fig. 4. WCOJ GenericJoin execution: (a) number of tuple-level iterations and (b) wall-clock time. LpH-CEG ordering reduces both by 2.3×/2.6× on heterogeneous Hetionet joins, while matching max-degree on homogeneous data.

for LpBound and 1.21× for exact counts—differences among the last three within noise. None competes with DuckDB's optimizer here (0.43× median, at most 2.0×): IMDb's cycles do not trigger its tail, and on the other 7 queries almost every forced left-deep order exceeds 60 s while DuckDB finishes, so excluding them favors the forced orders. A better bound improves *which* left-deep plan is chosen, not whether one should be.

This positions LpH-CEG honestly: it is not a replacement for a well-tuned sampling optimizer, but a guide that bounds the worst case on the dense cyclic queries that defeat sampling—on our synthetic cliques, not on IMDb—consistent with the pessimistic-estimation rationale for worst-case-optimal evaluation [16].

H. JOB, TPC-H, Data Size and Skew (bs=1)

On the JOB-medium subset (CI: 3K, MK: 2K, MC: 1K rows) the two co-actor 4-cycles improve 1.94× (with keywords) and 2.80× (with companies) at bs=1 with weights w=0.5, and a star query by exactly 1.00×; the IMDb suite of Section VI-D is the systematic version. On TPC-H (SF=0.01) every query yields 1.00×: dimension keys such as `nation_id` (25 values) are near-uniform, and LpH-CEG needs both *cyclic structure* and *degree skew*. On synthetic cycle-4 (10 seeds) the improvement grows with data size, from 5.9±0.7× at 100 vertices to 33.9±1.5× at 2K (relative std <12%; a single-run sweep reaches 57× at 5K), and peaks at moderate skew: 2.3±0.5× at α=0.5, 16.1±0.8× at α=2.0, 6.6±0.4× at α=3.0. At low skew all ℓ_p-norms nearly coincide, at extreme skew ℓ_∞ is already near-tight, and in between—the range real social and knowledge graphs occupy [17]—ℓ_2 captures the heavy tail.

I. When LpH-CEG Does Not Help

LpH-CEG yields no improvement (1.00×) on (1) acyclic queries—no redundant coverage—and (2) cycles of replicated relations with near-symmetric sampled degrees (the 5K-edge Epinions cycles, Section VI-D); case (2) is a property of the sample, as 20K edges of the same graph give 2.06× on cycle-4 (Section VI-D). What is required is *asymmetric degree structure in cyclic joins*. Synthetic power-law graphs improve more (up to 614× at cycle-8) than real data, where the median over the long suites is 1.8–8.9× (tails 157× on Hetionet, 34× on IMDb): Zipf skew at $\alpha \approx 2$ exceeds real sub-linear degree distributions and one α is shared by all relations.

VII. Related Work

a) Block decomposition methods.: CEG [6] models pessimistic estimation as a shortest path over single-relation max-degree weights; multi-way block extensions [5] generalize this to groups of relations via hyperpaths. Both use only ℓ_∞ statistics. LpH-CEG replaces the scalar weights with LP-optimized ℓ_p-norm products and characterizes *when* the LP can improve: cyclic queries with redundant coverage (Theorem 6).

b) Information-theoretic bounds.: The AGM bound [1] uses fractional edge covers with relation cardinalities (ℓ_1-norms). PANDA [2] and MOLP [3] add max-degree constraints (ℓ_∞-norms). LpBound [4] generalizes to all ℓ_p-norms, showing that intermediate norms capture degree skew that max-degree misses. Two results since then bear directly on our reading. Im et al. [10] compute the polymatroid bound for simple degree constraints by a polynomial flow LP and prove the non-simple case ($|X|\leq 2$, $|Y|\leq 3$, the shape of a bs=2 block constraint) hard, which is why our flow bound over blocks fixes an attribute order; CorrBound [15] adds inner products of degree vectors across join columns—a two-relation statistic in our terms—to the entropic program and reports 6.7–16× over LpBound on cyclic queries. The first tightens inference at fixed statistics, the second enriches statistics at fixed inference. The ℓ_1-norm of a bs=2 block sequence *is* CorrBound's inner product (the join size). In a proxy that adds exactly those sizes to the entropic program, they are worth a median 31× over LpBound on our long suite, and the interface-conditioned block statistics a further 63× under the same program. The shared-interface block norms $\sum_b \deg_R(b)^p \deg_S(b)^p$ are ℓ_p-weighted inner products and may fall within CorrBound's

vocabulary; the one-side and common-neighbour statistics do not, since they depend on *which* values co-occur, not only how often—and the latter needs the join that CorrBound's sketches avoid (Section VI-C). Lin and Wang [18] likewise tighten degree-sequence bounds by counting claw *pairs*, a richer statistic again. SafeBound [19] deploys degree-sequence bounds with predicates on relational workloads, following pessimistic CE [16]; we target graph patterns and a per-step decomposition.

The entropy framework of Abo Khamis et al. [9], under which ℓ_p-norm constraints are information inequalities, is what Theorem 4 applies to block-joined relations.

c) Summaries and learned estimators.: COLOR [13] and PathCE [20] build compact summaries; GNCE [14], SPACE [21] and FactorJoin [22] learn estimates. All produce point estimates without a guarantee, for which pessimistic bounds are the safety fallback [5].

d) WCOJ algorithms.: Yannakakis+ [23] brings worst-case optimal joins to DuckDB and PostgreSQL, a natural integration point for our ordering; CompactLTJ [24] shrinks Leapfrog Triejoin; Capelli et al. [25] call for exactly the recursive upper bounds our residuals provide—though our 66-query study shows they do not beat max-degree ordering in general.

VIII. Practical Considerations

a) When to use LpH-CEG.: Use it on cyclic queries with $k \geq 4$ and moderately skewed data, with $\mathcal{P} = \{1, 2, \infty\}$; without a cycle, skip the LP. Degree sequences can come from catalog histograms ($\|\mathbf{d}\|_p = (\sum_i c_i d_i^p)^{1/p}$ over buckets (d_i, c_i)) at `ANALYZE` time, on the filtered relation under predicates. Optimizing $K{=}10$ candidate hyperpaths instead of one adds 12.7–14.3% on cycle-4 and 0% on cycle-5.

b) Which block size to pay for.: At $bs{=}1$, collecting degree sequences costs 22–27 ms against $\sim$1.9 ms to solve. $bs{\geq}2$ multiplies that term by a median $45\times$ ($24\times$ more entries, 9–19 MiB against $\sim$0.3 MiB), once per database; cataloged, a $bs{=}2$ estimate then costs 10 ms (Table IV), and all but one statistic per binary block need no join (Section VI-C). Use $bs{=}1$ when the bound is re-solved many times per statistics build (per candidate order, per prefix) and $bs{\geq}2$ when the pair joins can be materialized ahead of the workload (the ℓ_2 codegree norm needs them), solved by the flow LP (Section VI-E) unless per-step residuals are wanted.

IX. Conclusion

Replacing a max-degree product with a fractional cover LP is worth a great deal—$654\times$ at cycle-8—but on single-relation statistics little is left after it: on a fixed hyperpath the cover LP is optimal over every valid weight vector, yet sits 1.25–$1.43\times$ (synthetic) and a median 2.5–$4.3\times$ (real) above an entropic optimum that LpBound's flow LP already reaches in polynomial time. The headroom is in the statistics: block statistics are worth a median $56\times$ on short Hetionet cycles, 80–$2{,}500\times$ on longer ones and 4–$43\times$ on IMDb (17–$177\times$ with the flow LP), below the entropic LpBound on all 349 cycle queries over real data and $12/12$ synthetic instances. Richer statistics can reopen the program—on Hetionet and synthetic data the optimum beats the cover LP by 1.5–$5.1\times$ on them, on IMDb only at scale ($6.8\times$ at 1,000 films)—and a flow LP reaches that optimum in polynomial time on synthetic even cycles and $189/240$ real-data queries.

We also state the limits. The separation is regime-dependent: on homogeneous synthetic cycles the two axes are comparable; heterogeneous real data splits them, and our evidence is binary graph patterns on two real datasets (Hetionet, IMDb). Blocks cost a median $45\times$ more to collect, once per database, and the one statistic that needs the join is the one that matters; 30% of the short Hetionet suite gains nothing at $bs{=}1$; the residual ordering heuristic does not generalize; and upper bounds bias plans conservative, so pairing them with optimistic estimators is the natural next step. Even so, the question to ask of a pessimistic estimator may be less which inequalities it uses than which joins it will pay to materialize—and only then which program it runs on them. We offer this as a characterization of where bound headroom lies, not as a deployable optimizer: our queries are cycle patterns over real data, of the kind subgraph-counting work targets [13], [20], our statistics come from slices (20K rows, 200–1,000 films) collected by a Python prototype whose cost grew $11\times$ for $4.2\times$ the data, and full-scale use needs the statistics cataloged inside the system.

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Use of AI-generated content. In accordance with the IEEE policy on AI-generated content, we disclose that Claude (Anthropic), a large language model used through the Claude Code tool, was used throughout this work. It drafted and revised text in every section (Sections I–IX and the abstract); implemented and ran the experiment scripts behind Sections VI and V and all tables; wrote the first version of the proof of Theorem 4; and produced simulated reviews of drafts. Every number in the paper is produced by the released scripts and checked against its source data by an automated audit. The authors reviewed the content and take full responsibility for it.

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